

Harshit Gupta

📞 +91 8340558673 | ✉️ harshitgupta.gh53@gmail.com | 🌐 harshit053 | 🎮 harshit053 | 📍 Bangalore, India

SUMMARY

GCP-native Data Engineer with 3.9 years of experience designing, optimizing, and delivering large-scale ETL/ELT pipelines on Google Cloud Platform. Specialized in building high-performance **PySpark** applications, cost-efficient **BigQuery** solutions, and robust data orchestration using **Apache Airflow** at American Express.

SKILLS

Languages and Frameworks: Python3, PySpark, SQL (Spark SQL), Apache Spark, Apache Airflow, BigQuery, pytest.

Data Engineering and Cloud: ETL/ELT, GCP – BigQuery, Dataproc, Cloud Composer, Pub/Sub, Dataflow, Apache Beam, Databricks, Snowflake, Parquet, JSON, CSV.

AI Assisted Development: GitHub Copilot, Claude Code, ChatGPT, Google Agent Development Kit, Model Context Protocol.

Secondary Tech Stack: HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Bootstrap.

Additional Skills: Data Structures and Algorithms, Git, GitHub, Agile Methodology (Jira, Rally).

CERTIFICATIONS

Google Cloud Platform – Data Engineer, Big Data

WORK EXPERIENCE

Infosys Ltd

Bangalore, India

Client: American Express – Engineer II (Data Engineer)

August 2022 – Present

- Engineered robust **PySpark** ETL pipelines on **Dataproc** to process and migrate complex multi-partition datasets from **BigQuery** to enterprise mainframe systems utilizing **complex analytical queries** for downstream transformations.
- Designed and maintained end-to-end enterprise data solutions by architecting data pipelines across the full **GCP** data ecosystem, seamlessly integrating **BigQuery** for warehousing, **Dataproc** for Spark processing, **Cloud Composer** for orchestration, **Cloud Storage** for data lake, **Pub/Sub** for messaging, **Dataflow** for ingestion, and **Cloud Functions** for event triggers.
- Architected **BigQuery** storage patterns using **table partitioning** and **clustering** to minimize data scans and reduce slot consumption, **lowering query costs by ~20%** across large enterprise datasets.
- Automated and monitored **40+ Airflow DAGs** in **GCP Cloud Composer** processing **~1 TB of data daily**; managed complex **task dependencies** and integrated **Cloud Pub/Sub** for event-driven pipeline triggering.
- Tuned **Spark** performance on **Dataproc** via **broadcast joins**, **data caching**, **skew mitigation**, **dynamic allocation**, and **ephemeral clusters**, **reducing execution time by ~30%**.
- Handled data ingestion using industry-standard file formats including **Parquet** and **JSON** for structured data exchange between **GCS**, **BigQuery**, and downstream mainframe systems, ensuring schema compatibility and efficient serialization.
- Developed **Python**-based SFTP scripts with **PGP Encryption** to securely transfer files from **GCS** to mainframe; wrote **pytest** unit tests and **BDD** (Gherkin) scenarios to ensure pipeline reliability.
- Leveraged **Agile** methodologies (Rally, Jira) to collaborate with cross-functional teams, manage sprint deliverables, and align pipeline development with business requirements.

PROJECTS

Data Processing Pipeline | *Python3, PySpark, Apache Spark, SQLite*

Website | GitHub

- Demonstrated a simple data processing pipeline designed to clean, transform, and analyze data.
- Optimized join operations using **broadcast joins** to efficiently combine resulting Spark DataFrames.
- Validated records by identifying suspect records and separated good records from bad ones.
- Created an appropriate data model in an **SQLite** database and ingested the cleaned data into it.

Skills Learned: Data extraction, Data handling and manipulation, Data ingestion

YelpCamp | *MongoDB, Express.js, EJS, Node.js, Passport.js, Bootstrap*

Website | GitHub

- Developed a full-stack campground review site with user accounts, campground listings, and review posting.
- Built complete **CRUD** functionality for campgrounds and reviews, adhering to **RESTful API** principles.

Skills Learned: Full-Stack Development, CRUD operations, RESTful API Principles

DigiMenu | *HTML, CSS, JavaScript, MongoDB, Node.js, Express.js, Passport.js*

Website | GitHub

- Built a digital menu application allowing users to browse and view available items and categories.
- Implemented **authentication and authorization** using **Passport.js** with session-based login.

Skills Learned: CRUD operations, RESTful routing, Authentication and Authorization

EDUCATION

Visvesvaraya Technological University

Bangalore, Karnataka, IN

B.E. in Electronics and Communication | Overall: 8.8 CGPA

August 2018 – July 2022